



Entergy Nuclear Operations, Inc.
Pilgrim Nuclear Power Station
600 Rocky Hill Road
Plymouth, MA 02360

Peter J. Miner
Manager, Regulatory Assurance

Letter Number 2.18.002

October 25, 2018

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

SUBJECT: Licensee Event Report 2017-012-01, Supplement to Start-Up Transformer
Degraded Voltage Relay Found Outside Technical Specification Limit

Pilgrim Nuclear Power Station
Docket No. 50-293
Renewed License No. DPR-35

Dear Sir or Madam:

The enclosed Licensee Event Report 2017-012-01, Supplement to Start-Up Transformer Degraded Voltage Relay Found Outside Technical Specification Limit, is submitted in accordance with Title 10 Code of Federal Regulations 50.73.

If you have any questions or require additional information, please contact me at (508) 830-7127.

There are no regulatory commitments contained in this letter.

Sincerely,

A handwritten signature in black ink, appearing to read "PJM", with a long horizontal flourish extending to the right.

PJM/rm

Attachment 1: Licensee Event Report 2017-012-01, Supplement to Start-Up Transformer
Degraded Voltage Relay Found Outside Technical Specification Limit

TEZZ
NRR

cc: Mr. David C. Lew
Regional Administrator, Region I
U. S. Nuclear Regulatory Commission
2100 Renaissance Boulevard, Suite 100
King of Prussia, PA 19406-2713

Mr. John Lamb, Senior Project Manager
Office of Nuclear Reactor Regulation
U.S. Nuclear Regulatory Commission
Mail Stop 0-8C2A
Washington, DC 20555

NRC Senior Resident Inspector
Pilgrim Nuclear Power Station

Attachment 1

Letter Number 2.18.002

Licensee Event Report 2017-012-01, Supplement to Start-Up Transformer Degraded
Voltage Relay Found Outside Technical Specification Limit
(3 Pages)



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME Pilgrim Nuclear Power Station	2. DOCKET NUMBER 05000-293	3. PAGE 1 OF 3
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4. TITLE Supplement to Start-Up Transformer Degraded Voltage Relay Found Outside Technical Specification Limit

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
09	13	2017	2017	- 012	- 01	10	25	2018	N/A	05000 N/A
									FACILITY NAME	DOCKET NUMBER
									N/A	05000 N/A

9. OPERATING MODE N	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
10. POWER LEVEL 100	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT Mr. Peter J. Miner - Regulatory Assurance Manager	TELEPHONE NUMBER (Include Area Code) 508-830-7127
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13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX
B	EA	27	ABB	Y	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED

☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE)☒ NO

15. EXPECTED SUBMISSION DATE

MONTH DAY YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On September 13, 2017 with the unit at 100 percent power, Pilgrim Nuclear Power Station discovered during calibration of four Start-Up Transformer Degraded Voltage Relays (ABB model number ITE-27N) after the relays had been removed from the plant, relay 127A-604/1 had an as-found setpoint value outside the Technical Specifications Table 3.2.B limit of 110.6 plus or minus 0.56 Volts ac (Vac) (110.04 – 111.16); the as-found value was 109.90 Vac. This relay was replaced with a spare relay that was calibrated prior to its installation.

Pilgrim Nuclear Power Station is submitting this Licensee Event Report in accordance with 10 CFR 50.73(a)(2)(i)(B) – Any operation or condition that was prohibited by the plant's Technical Specifications.

This event was not risk significant. There was no threat to public health and safety from this condition.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Pilgrim Nuclear Power Station	05000-293	2017	- 012	- 01

NARRATIVE

BACKGROUND

The startup transformer supplies power to the station auxiliary power distribution system whenever the main generator is offline. After the main generator has been synchronized to the 345 kV system and has been partially loaded, the auxiliary power distribution system is manually transferred from the start-up transformer (SUT) to the unit AC power source (unit auxiliary transformer [UAT]).

Automatic fast transfer capability is provided in the design to restore the auxiliary power distribution system to its preferred offsite AC power source, the SUT, in the event that the unit AC power source (UAT) is lost for any reason. The diesel generator load shedding logic will also be actuated immediately upon the fast transfer of the safety related buses A5 and A6 to the SUT if a Loss-Of-Coolant-Accident signal is present when the SUT secondary voltage is below the degraded voltage alarm reset setpoint.

The primary function of the 127A-604/1 degraded voltage relays and their settings for the SUT is to ensure a minimum voltage at which the emergency buses can be operated indefinitely to provide their intended safety functions. If voltage drops below that point for a specified time limit, the degraded voltage relays will initiate transfer of the emergency buses from offsite power to the onsite emergency diesel generators.

EVENT DESCRIPTION

During calibration of four SUT Degraded Voltage Relays (ABB model number ITE-27N) after the relays had been removed from the plant, relay 127A-604/1 had an as-found setpoint value outside the Technical Specifications (TS) Table 3.2.B limit of 110.6 plus or minus 0.56 Volts ac (Vac) (110.04 – 111.16). The as-found value was 109.90 Vac. This relay was replaced with a spare relay that was calibrated prior to its installation.

CAUSE OF THE EVENT

A preliminary failure analysis concluded that the relay was exhibiting drift. However, testing at a vendor facility concluded that the relay had repeatable good performance with no indication of unacceptable drift. Similarly, other identical relays purchased at the same time did not show signs of excessive drift. Therefore, the cause of the relay failure remains inconclusive. However, the failure was unique to the affected relay, which was replaced. Based on a review of calculations, margins exist to compensate for the minor deviation, which indicates that the degraded voltage relay would operate within safety limits but outside the TS limits.

CORRECTIVE ACTIONS

The relay was replaced, restoring the safety function of the SUT degraded voltage relay to initiate transfer



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CONTINUATION SHEET**

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Pilgrim Nuclear Power Station	05000-293	YEAR 2017	SEQUENTIAL NUMBER - 012	REV NO. - 01

NARRATIVE

of the emergency buses from offsite power to the onsite emergency diesel generators.

SAFETY CONSEQUENCES

The 127A-604/1 Degraded Voltage Relay As-Found setpoint value was outside the TS limits. However, per calculation PS147 "Degraded Voltage Trip Relays, Revised Voltage Setpoint" there was sufficient margin, which indicates that the degraded voltage relay would operate within safety limits but outside the TS limits.

Calculation PS147 "Degraded Voltage Trip Relays, Revised Voltage Setpoint" establishes the trip setpoint, analytical limits, and the no adjust limits for the voltage function of the degraded voltage trip relays (127A-504/1,2,3,4 and 127A-604/1,2,3,4). The upper and lower analytical limits are the limit of a measured or calculated variable established by the plant safety analysis to ensure that a safety limit is not exceeded. The as-found value found for this relay was 109.9 Vac which was outside the TS limits. However, per PS147, adequate margin exists in electrical load flow calculations and instrument variable uncertainties used in Total Loop Uncertainties, which ensures that the lower analytical limit was not exceeded. Therefore, the as-found value for 127A-604/1 does not exceed the established safety limit, and the relay is capable of performing its safety function.

There were no public safety, nuclear safety, industrial safety or radiological safety consequences from this event. No actions to reduce the frequency or consequence are necessary.

REPORTABILITY

Pilgrim Nuclear Power Station is submitting this Licensee Event Report (LER) in accordance with 10 CFR 50.73(a)(2)(i)(B) – Any operation or condition that was prohibited by the plant's Technical Specifications.

PREVIOUS EVENTS

A review of PNPS LERs for the past five years did not identify any other LERs that were submitted for the same reason as this submittal.

REFERENCES:

CR-PNP-2017-9211
CR-PNP-2018-5532